



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Spring 2025

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors.

[8]

```
1. include <studio.h>
2. include<String.h>
3. #define PI = 3.14159:
4. double calculateTemp(double t):
5. int lengthCount (char x[], );
6. int Main() {
7. char string[] ;
8. print('Enter String: /n');
9. Gets(string,10);
10. int a[5] = {1,2,3,4,5,6};
11. for(i = 0: i>5;i++){
12.     sum = + a[j] :
13. }}
14. fi(sam<5)
15.     print("sum is greater then 5",);
16. printf('\n');
17. printf("%s" strang);
18. String y[5];
19. strcpy(yy, string);
20. ("%d", calculateTamp(sum));
21. printf("%s", lengthCount (string, 5));
22. return 0;

23. double calculateTemp (int t, int p){
24.     t =/ 2.1;
25.     t = t/2+ +10:
26.     return 10;
27. }

28. float lengthCounts (char y){
29.     int len= -1;
30.     whale(y[len] != "0" ){
31.         lans- -;
32. }
```

2. Construct a Non-recursive Predictive Parsing table for the following grammar & also check whether string "3333452p" is accepted or not. The sequence of Moves by Predictive Parser in a table (should have column: stack,input, action).

[4]

```
S -> A
A -> BCD | BC
B -> 3B | ε
C -> 1 | 4C | 5C | ε
D -> 2D | p
```

3. For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. Consider all the terminals are of single character/number/symbol. [4*3]

- a) $A \rightarrow aBi \mid bBCDiu \mid BqCr \mid BCtu \mid \epsilon$
 $B \rightarrow 1BAi \mid BCDu \mid B3ApB \mid CD2 \mid \epsilon$
 $C \rightarrow CrDt \mid zDBl \mid BbCy2 \mid iABcDkB \mid \epsilon$
 $D \rightarrow 0D1B \mid Cx2 \mid DD2Ai \mid 3B2ACi \mid \epsilon$
- b) $X \rightarrow XaYkZ \mid XbGyi \mid 22Xgk \mid \epsilon$
 $Y \rightarrow 3GGZ \mid YYiZm \mid qtXGt \mid GkXX \mid \epsilon$
 $Z \rightarrow yXgYZ \mid GrYgZp \mid YoYGk \mid tGn \mid \epsilon$
 $G \rightarrow tGX3 \mid GkBk \mid X3GXi \mid 4Zi \mid \epsilon$
- c) $P \rightarrow bPQc \mid RPPy \mid xQrRg \mid \epsilon$
 $Q \rightarrow 1QRb \mid RR25 \mid P2PlR \mid \epsilon$
 $R \rightarrow 3RQk \mid 5tQP \mid 6RRyi \mid \epsilon$

4. Left Factor the following. [4*2]

- (a) $P \rightarrow (2(2+3)) \mid (2+b) \mid (2(+3(4))) \mid (2-3(5)) \mid (2-3) \mid (2+b(1*2)) \mid (2(+2(1/2))) \mid (a) \mid (2(2-6)) \mid (2+4) \mid (2(3+5)) \mid (2(+3(1/2))) \mid (2(+2(1*2))) \mid (2(+2(1/5)))$
- (b) $R \rightarrow \text{Rabit Laughs} \mid \text{Rabit Loves Radish} \mid \text{Rabits talks} \mid \text{Rabit Loving} \mid \text{refrening} \mid \text{restrained} \mid \text{referee} \mid \text{reference} \mid \text{Rabits} \mid \text{Rabit Lashes} \mid \text{refer} \mid \text{Rabit Lost} \mid \text{Rabit Loving} \mid \text{Rabit Lasting} \mid \text{Rab} \mid \text{Rabs} \mid \text{Ra pitch}$

5. Eliminate immediate left recursion from the following grammar. [4*2]

- (a) $P \rightarrow PP+RsT \mid (PabRT) \mid PghQr \mid P^*+QQ$
 $Q \rightarrow Q2Ph \mid QPiQ \mid RaQ \mid RcbQQ \mid \epsilon$
 $R \rightarrow R(2+T) \mid RciTS \mid aRsTP \mid R \mid \epsilon \rightarrow SSap \mid SP/T \mid S \mid d/SC \mid abSgt$
 $T \rightarrow (RT) \mid Ta \mid bT \mid Tc \mid TTE$
- (b) $X \rightarrow Xo[4]Y \mid XX*ZI \mid Z+YY \mid \epsilon$
 $Y \rightarrow Y+2[3]Zi+ht \mid IY/2f \mid Y2*ab \mid \epsilon$
 $Z \rightarrow ZOX \mid oZY \mid Zoo \mid ZYot \mid \epsilon$
 $I \rightarrow 1 \mid 2 \mid 3 \mid$